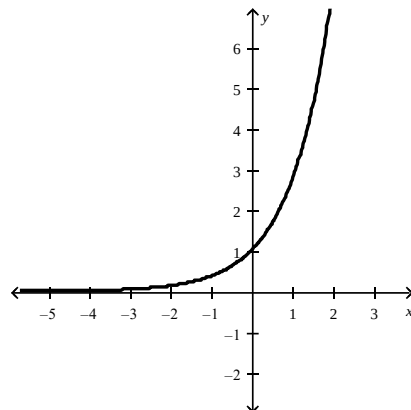


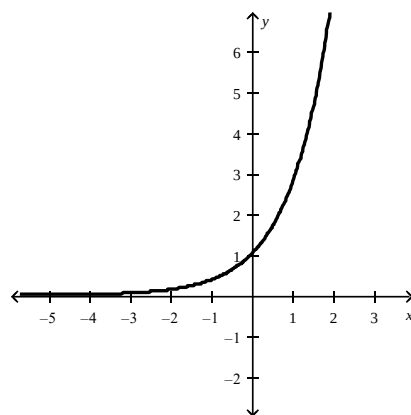
AP Calculus II
Notes 9.7
Taylor and Maclaurin Polynomials

The purpose of this next unit is to show how transcendental and trigonometric functions can be approximated by polynomials.

Let's start off with trying to approximate the curve $f(x) = e^x$ at $x = 0$. The most basic approximation of this function would have a degree of 0 (a constant) and would be:

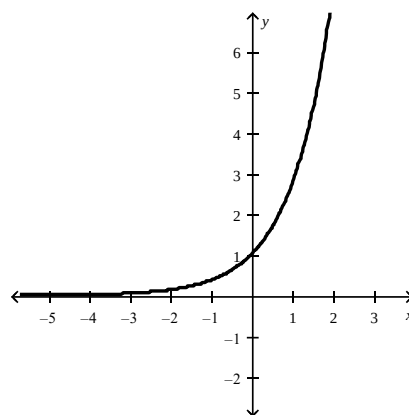


How can we make this approximation better?



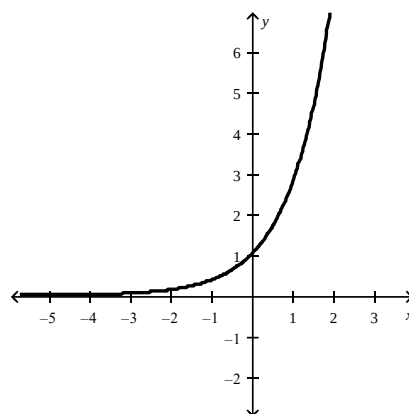
Check:

How can we make *this* approximation any better?



Check:

How can we make this approximation any better?



Check:

Definition of Maclaurin Polynomials:

If $f(x)$ has n derivatives at $c = 0$, then:

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

is also known as the **nth Maclaurin polynomial for $f(x)$** that is centered at $c = 0$.

Ex. 1: Find the Maclaurin polynomials P_4, P_5 and P_6 for $f(x) = e^x$ and use each to approximate $e^{0.1}$.

Ex. 2: Find the Maclaurin polynomial of degree 6 for $f(x) = \cos 2x$. How many nonzero terms are in the polynomial?

Ex. 3: Find the first 4 nonzero terms of the Maclaurin polynomial for $g(x) = \frac{6}{2+x}$.

Ex. 4: Find the Maclaurin polynomial of degree 6 for $f(x) = \cos x + \sin x$. How many nonzero terms are in the polynomial?

Definition of Taylor Polynomials:

If at c , then:

$$P_n(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \frac{f'''(c)}{3!}(x - c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n$$

is also known as the **nth Taylor polynomial for $f(x)$** . A Maclaurin Polynomial is a specific type of Taylor Polynomial with $c = 0$.

Ex. 5: Find the third-degree Taylor polynomial for $f(x) = 2 \sin x$, centered at the following values:

a) $c = 0$

b) $c = \frac{\pi}{6}$

c) $c = \frac{3\pi}{4}$

Ex. 4: Find the Taylor polynomial P_3 for $f(x) = \ln|\sin x|$ at $c = \frac{\pi}{2}$. Then, use P_3 to approximate $f(1.5)$.

Ex. 5: Find the fourth degree term of the Taylor polynomial for $f(x) = x^2 + \ln x$ centered at $c = 1$.

Ex. 6: Let f be a function such that $f(2) = 4$, $f'(2) = -3$ and $f^{(n+1)}(2) = -2f^{(n)}(2)$, for all $n \geq 1$.

a) Write the first four nonzero terms of the Taylor Polynomial for f about $x = 2$.

b) Write the fourth degree Taylor Polynomial for g , where $g(x) = f'(x)$ about $x = 2$.

c) Write the third degree Taylor Polynomial for h , where $h(x) = x^3 + \int_2^{3x-4} f(t)dt$ about $x = 2$.

Ex. 7: The third-degree Taylor polynomial for the function f about $x = 0$ is $T(x) = 3 - 4x + 2x^2 - 3x^3$. Which of the following tables gives the values of f and its first three derivatives at $x = 0$?

(a)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-8	6	-12

(b)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-4	2	-3

(c)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-4	4	-18

(d)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-4	4	-9

Ex. 8: The functions f , f' , and f'' are each continuous and differentiable. The n th derivative of f is given by $f^{(n)}(1) = \sum_{i=0}^{\infty} 12 \left(\frac{n+1}{5} \right)^i$ when $0 \leq n \leq 3$. Find the third degree Taylor polynomial for $f(x)$ centered around $x = 1$.

AP Calculus II
Notes 9.10
Taylor and Maclaurin Power Series

In the last section, we found that Taylor and Maclaurin polynomials can approximate various functions. For example, we know the Maclaurin polynomials $P_1 \rightarrow P_5$ for the function $f(x) = e^x$ are as follows:

We also found that the higher the degree of the polynomial, the better the approximation becomes about c . Therefore, if we wanted to find the *exact* representation of these functions, we use **power series**.

Definition of Power Series

If x is a variable, then an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots + a_n(x-c)^n$$

is called a **power series centered at c** and $a_n(x-c)^n$ is called the **general term**.

Taylor and Maclaurin Power Series

If a function $f(x)$ has derivatives of all orders at $x = c$, then the series:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!} (x-c)^2 + \frac{f'''(c)}{3!} (x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!} (x-c)^n$$

is called the **Taylor series for $f(x)$ centered at c** . Again, a Taylor series centered at $c = 0$ is called a **Maclaurin series**.

If you can find the pattern for the coefficients of the Taylor polynomials for a function, then you can extend the pattern easily for the corresponding Taylor series.

Ex. 1: Write the fourth-degree Maclaurin polynomial for $f(x) = e^x$, write the general term and the Maclaurin series for $f(x) = e^x$.

Ex. 2: Write the fourth-degree Taylor polynomial for $f(x) = \ln x$, centered about $c = 1$, write the general term and the Taylor series for $f(x) = \ln x$, centered about $c = 1$.

Ex. 3: Write the Maclaurin series for $f(x) = \sin x$.

Ex. 4: Write the Maclaurin series for $f(x) = \cos x$.

The Maclaurin Series You MUST Know:

1)

2)

3)

4)

Ex. 5: Find the Maclaurin power series for $f(x) = \sin(x^2)$.

Ex. 6: Find the Taylor power series for $f(x) = \frac{e^{(-3x)} - 1}{3}$ centered about $x = 0$.

Ex. 7: Find the first three nonzero terms in the Maclaurin series $g(x) = x \cos x - \sin x$.

Ex. 8: Evaluate the following infinite series:

a)
$$\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{3^{2n} (2n)!}$$

b)
$$4 - \frac{4x^2}{3!} + \frac{4x^4}{5!} - \frac{4x^6}{7!} + \dots$$

c)
$$1 - 2 + \frac{4}{2!} - \frac{8}{3!} + \frac{16}{4!} - \frac{32}{5!} + \dots$$

d)
$$1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \dots$$

Ex. 9: Given $f(x) = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \cdots + \frac{(-1)^n 2x^{2n+1}}{(2n+1)!}$,

- a) Find one value of x for which $f(x) = 1$.
- b) Find the first 4 non-zero terms for $f'(x)$. Write the Maclaurin series for $f'(x)$?
- c) If $g'(x) = f(x)$ and $g(0) = -2$, find the first 4 non-zero terms and the general term of $g(x)$.
- d) Find $f^{(19)}(0)$.

AP Calculus II
Notes 9.8
Power Series

In the last section, we dealt with Maclaurin and Taylor polynomials that when added up, resulted in Maclaurin and Taylor series. In this section, we will focus on any power series and its features.

Definition of Power Series

If x is a variable, then an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x - c)^n = a_0 + a_1(x - c) + a_2(x - c)^2 + a_3(x - c)^3 + \cdots + a_n(x - c)^n$$

is called a **power series centered at c** .

When representing a function $f(x)$ as a power series, the **domain** of the function $f(x)$ is all the values of x for which the power series converges. The Geometric, Alternating Series and Ratio Tests are very helpful in these situations, but the other tests are just as useful. The set of all x –values for which the power series converges is known as the **interval of convergence**.

Interval of Convergence of a Power Series:

For a power series centered at c , one of the following cases is true, along with a graphical representation:

- a) The series converges for all x . ($R = \infty$)

- b) The series converges for only a finite window $|x - c| < R$, with c being the center of this window and R being the **radius of convergence**. Think of R as being the “plus or minus” away from the center. This interval of convergence can include one, none or both endpoints.

- c) The series converges at one point, namely c . ($R = 0$)

Ex. 1: Find the center, radius and interval of convergence for the following power series:

a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

b) $\sum_{n=1}^{\infty} n! x^n$

c) $\sum_{n=3}^{\infty} (-1)^{n+1} \frac{(x+3)^n}{(n-2)^3}$

Ex. 2: Write out the first 4 nonzero terms and the general term of the Maclaurin series for $f(x) = \sin x$. Then, determine the interval of convergence for this function.

Ex. 3: The Taylor series for the function $f(x) = \ln x$ is $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(x-1)^n}{n}$. Find the interval of convergence for this function. Then, find the value of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.

Ex. 4: The series $\sum_{n=1}^{\infty} a_n (x-2)^n$ converges conditionally when $x = 0$. Determine if the following statements are **Always true**, **Sometimes true** or **Never true**.

a.) The series diverges when $x = 5$.	b.) The series converges when $x = -1$.
c.) The series diverges when $x = 1$.	d.) The series converges conditionally when $x = 4$.

Ex. 5:
$$\sum_{n=1}^{\infty} b_n = -\frac{4}{3} + \frac{4 \cdot 2}{9}(x-2) - \frac{4 \cdot 3}{27}(x-2)^3 + \cdots + (-1)^n \frac{4n}{3^n}(x-2)^{n-1} + \cdots$$

Find the interval of convergence for the series $\sum_{n=1}^{\infty} \frac{b_n}{n^2}$.

Power Series Manipulation

Since these series represent different functions, then we can manipulate the series the same way we manipulate functions. This allows us to differentiate, integrate, find extrema, points of inflection, etc...

Ex. 6: Given $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$, find the following:

a) The first four terms and the general term for $f'(x)$. Then, find $f'(0)$ and $f^{(11)}(0)$.

b) The first four terms and the general term for $\int f(x)dx$.

c) The first four terms and the general term for $g'(t)$ given $g(t) = f(t^3)$.

Ex. 7: Given $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^{n+1}}{(n+1)(n+2)}$, find the interval of convergence for $f'(x)$.

Ex. 8: Find the first 3 nonzero terms and the general term for $\int_0^x \sin(t^4) dt$.

Ex. 9: Given $f(x) = \frac{\cos x - 1}{x^2}$, find the following using series:

a) $\lim_{x \rightarrow 0} f(x)$, then verify using L'Hôpital's rule.

b) First 4 nonzero terms and the general term for $\int f(x) dx$.

Ex. 10: Given $g(x) = \frac{e^x - e^{-x}}{2}$, find:

a) The power series expansion of $g(x)$ by showing the first 4 nonzero terms and general term.

b) The relationship between $g(x)$ and the series $h(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$

c) The equation represented in the power series expansion for the function $h(x)$.

Ex. 11:

The function f is defined by the power series

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots + \frac{(-1)^n x^{2n}}{(2n+1)!} + \cdots$$

for all real numbers x .

- (a) Find $f'(0)$ and $f''(0)$. Determine whether f has a local maximum, a local minimum, or neither at $x = 0$.
Give a reason for your answer.
- (c) Show that $y = f(x)$ is a solution to the differential equation $xy' + y = \cos x$.

Ex. 12: The function f has derivatives of all orders at $x = 0$, and the Maclaurin series for f is

$$\sum_{n=2}^{\infty} \frac{\ln n}{3^n n^3} x^n.$$

- (a) Find $f'(0)$ and $f^{(4)}(0)$.
- (b) Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer.
- (c) Using the ratio test, determine the interval of convergence of the Maclaurin series for f . Justify your answer.

AP Calculus II
Notes 9.9
Geometric Power Series

Consider the series $\sum_{n=0}^{\infty} x^n$. We know that this sum can be written as:

$$\sum_{n=0}^{\infty} x^n =$$

This expression closely resembles the sum of a geometric series with $a =$ and $r =$. Therefore,

the sum of the series would be: $\sum_{n=0}^{\infty} x^n =$

We can now express functions that look like the sum of a geometric sequence as a **geometric power series** using the general representation that has a center at c :

$$\sum_{n=0}^{\infty} a(x - c)^n =$$

Ex. 1: Write the following expressions as a geometric power series and then write the first 4 terms:

a) $\frac{12}{1 - \frac{1}{4}}$

b) $\frac{3}{1 - (x + 2)}$

c) $\frac{5x^2}{1 + x}$

Ex. 2: Find a power series for $f(x) = \frac{4}{x+5}$, centered at the following values. Then, find the interval of convergence for each power series.

a) $c = 0$

b) $c = 3$

Ex. 3: Evaluate $\int \frac{6x}{1-x^3} dx$ using power series centered at 0.

Ex. 4: If $h'(x) = 1 - 3x + 9x^2 - 27x^3 + \dots$

a) Write the function represented by $h'(x)$ and then write the Maclaurin series for $h(x)$.

b) Find the radius of convergence of $h(x)$.

Ex. 5:

The Maclaurin series for the function f is given by $f(x) = \sum_{n=2}^{\infty} \frac{(-1)^n (2x)^n}{n-1}$ on its interval of convergence.

- (a) Find the interval of convergence for the Maclaurin series of f . Justify your answer.
- (b) Show that $y = f(x)$ is a solution to the differential equation $xy' - y = \frac{4x^2}{1+2x}$ for $|x| < R$, where R is the radius of convergence from part (a).

AP Calculus II

Notes 9.7

Error Approximation

One of the bigger uses of Taylor polynomials is to approximate values of a transcendental/trigonometric function. This leads us to the idea of the accuracy of the approximation.

$$f(c) = P_n(c) + R_n(c)$$

So, $R_n(c) = f(c) - P_n(c)$. This means that the **error** of the approximation can be expressed as:

Error =

We can find this error using one of two methods, which depend on whether the polynomial alternates. We will start off with if the polynomial does alternate:

Alternating Series Remainder:

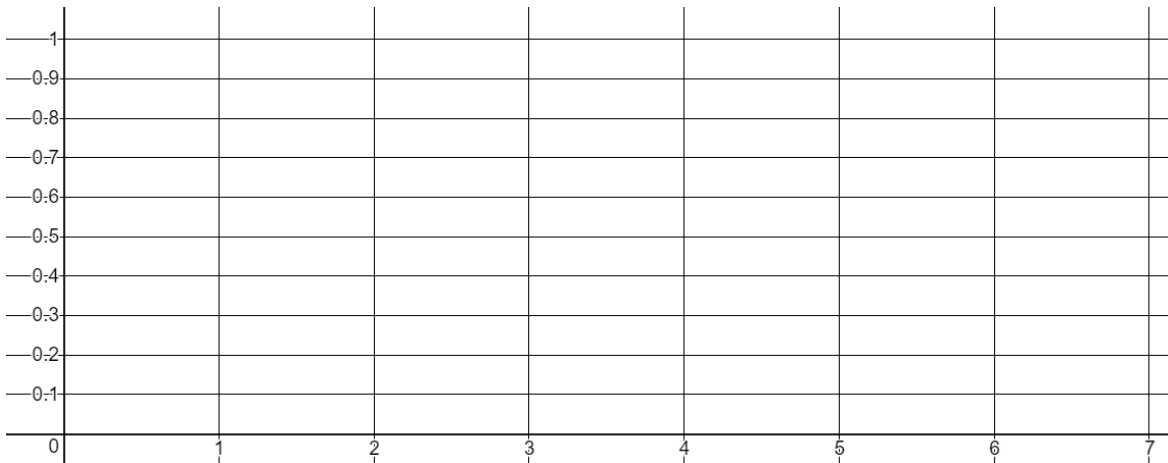
If a series $\sum_{n=1}^{\infty} a_n$ is **alternating, decreasing in absolute value** and $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.

If you look at the partial sums of the series, it will bounce back and forth around the actual sum. To model this, let's look at the following geometric series:

$$\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \dots$$

Knowing that this series is geometric, we know that this series adds up to _____.

Let's imagine we didn't know what this series equals, so let's approximate the sum using the first few partial sums.



So, if one partial sum is larger than the true value, then subtracting the next term will cause the partial sum to be smaller and thus, less than the true value. Then, adding the next term will cause the partial sum to be larger and so on... The error is the difference between any of the partial sums and the actual value, but by adding (or subtracting) the next term, the partial sum will go past the actual value. Therefore, the **maximum error** is less than or equal to the **first (next) neglected term** in the series.

Again, this applies to **convergent alternating series that decrease in absolute value to 0.**

Ex. 1: Given $g(x) = \frac{6}{2+x}$, write the first four nonzero terms for the Maclaurin polynomial and the general term for $g(x)$. Using this polynomial, approximate $g(1)$ and determine the maximum error.

Ex. 2: Given $f(x) = \sin x$, write the fifth degree Maclaurin polynomial and the general term for $f(x)$. Using this polynomial, approximate $\sin(0.1)$ and determine the maximum error.

Ex. 3: Given $f(x) = e^{-2x}$,

a) Write the third degree Maclaurin polynomial $P_3(x)$ for $f(x)$.

b) Using $P_3(x)$, approximate $f(0.5)$. Then, show that $f(0.5)$ differs from $P_3(0.5)$ by less than $\frac{1}{20}$.

Ex. 4: The Taylor series for f about $x = 2$ converges to f for $-1 \leq x \leq 5$. The n th-degree Taylor polynomial for f about $x = 2$ is given by $P_n(x) = \sum_{k=1}^n (-1)^k \frac{(x-2)^k}{3^k(k^2+1)}$. What is the smallest degree n of the Taylor polynomial that guarantees that $|f(3) - P_n(3)| \leq \frac{1}{100,000}$?

Ex. 5: The Taylor series about $x = 4$ for a certain function f converges to $f(x)$ for all x in the interval of convergence. The n th derivative of f at $x = 4$ is given by

$$f^{(n)}(4) = \frac{(-1)^n(n+1)!}{2^n} \quad \text{and } f(4) = 1$$

a) Write the third-degree Taylor polynomial for $f(x)$ centered at $x = 4$ and find the general term.

b) Find the coefficient of the 10th degree term of the Taylor series for f centered at $c = 4$.

c) Show that the third-degree Taylor polynomial approximates $f(5)$ with an error less than $\frac{1}{3}$.

What if the series is not alternating?

If the series is not alternating, you must use the Lagrange Error in which the maximum error will occur at some value z , such that z is in between x and c .

Lagrange Error Bound/Taylor's Theorem:

$$\text{For } f(x) = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + R_n(x)$$

$$\text{where the maximum value of the remainder is } R_n(x) = \left| \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1} \right|.$$

This is very similar to the Alternating Series Remainder in that they both use the next term in the series. However when using Lagrange Error, you must find or use reasoning to determine the z value that will make the next derivative the maximum value possible. This will make the remainder the largest possible and will truly represent the maximum error.

Ex. 6: Given the function $f(x) = 3e^x$,

- a) Write the fourth-degree Maclaurin polynomial for $f(x)$ and the general term. Use this fourth-degree polynomial to approximate $3e^2$.

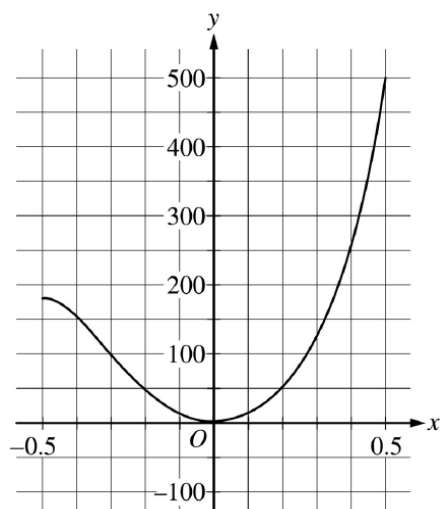
- b) Find the maximum error of this approximation to the actual value of $3e^2$.

Ex. 7: The function $f(x) = \frac{1}{(1-x)^2}$ can be expressed as the series $\sum_{n=1}^{\infty} nx^{n-1}$ for all values of x within its interval of convergence.

a) Find the interval of convergence of $f(x)$.

b) Use the first 4 nonzero terms of f to approximate the value of $f\left(\frac{1}{3}\right)$. Show that this approximation differs from $f\left(\frac{1}{3}\right)$ by less than $\frac{3}{4}$.

Ex. 8: Let the function f be defined as $f(x) = x^2 e^{x^2}$. The graph of $f^{(7)}$, the seventh derivative of f , can be seen below.



- a) Write the first four nonzero terms and general term for the Maclaurin series for e^x . Then, use this series to write the first four nonzero terms and general term for the Maclaurin series for f .

- b) Let $P_6(x)$ be the sixth-degree Taylor polynomial for f about $x = 0$. Find $P_6\left(\frac{1}{2}\right)$, then use the Lagrange error bound along with the graph to find an upper bound on $\left|P_6\left(\frac{1}{2}\right) - f\left(\frac{1}{2}\right)\right|$.